# Polynomial Approximation of the Riemann Ξ Function in the Log-Hermite Polynomials

Robin Langer and Claude Code

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**Abstract**

We approximate $\Xi(T) = \xi(\frac{1}{2} + iT)$ by *polynomials* — on $\mathbb{R}_+$, by $x^{-1/2}$ times polynomials in $\log x$ — rather than truncating its expansion in the log-Hermite basis functions, and compare the zeros of the two approximants. The projection coefficients obey the identity $\beta_k = \langle \Xi(\lambda\,\cdot)\, e^{-t^2/2},\, \psi_k \rangle$, so one Gaussian factor in the quadrature separates the two problems. The polynomial approximant is markedly worse at capturing Riemann zeros: at degree 60 it reproduces two (the sections reproduce five to six digits), because its tail modes grow like $e^{t^2/2}$ exactly where $\Xi$ decays. The crossover height is governed by a quadratic balance law in place of the sections' linear one, the measured coefficient decay is the inf-convolution of the arithmetic decay with the geometric rate $\frac{1}{2}\ln 3$ of the Hermite matching matrix, and the complex defects at the crossover now form short arcs. This adds a fourth row to the taxonomy of section-zero geometries.

## 1 Two approximations, one Gaussian apart

Let $\Xi(T) = \xi(\frac{1}{2} + iT)$ and $F(t) = \Xi(\lambda t)$ with $\lambda = 10$. Our companion paper studied the *sections* of the Hermite-function expansion $F = \sum_k b_k \psi_k$: truncations $S_N = \sum_{k \le N} b_k \psi_k$, each a polynomial times $e^{-t^2/2}$. Here we instead take the best approximation of $F$ by a *polynomial* of degree $\le N$ in $L^2(\mathbb{R}, e^{-t^2} dt)$:

$$P_N F \;=\; \sum_{k \le N} \beta_k\, \tilde{h}_k, \qquad \tilde{h}_k = \frac{H_k}{(2^k k! \sqrt{\pi})^{1/2}} = \psi_k\, e^{t^2/2}.$$

Under $x = e^t$ this is approximation by $x^{-1/2}$ times polynomials in $\log x$ against the log-normal weight: the *log-Hermite polynomial* approximation of the multiplicative side of $\zeta$. It is well posed — polynomials in $\log x$ transport to ordinary polynomials in $t$, where Hermite completeness holds — even though polynomials in $x$ are not dense against the log-normal weight (Stieltjes–Wigert indeterminacy).

The two computations differ by a single Gaussian:

$$\beta_k \;=\; \langle F, \tilde{h}_k \rangle_{e^{-t^2}} \;=\; \int_{\mathbb{R}} F(t)\, \psi_k(t)\, e^{-t^2/2}\, dt, \tag{1}$$

the Hermite-*function* coefficients of the damped target $F e^{-t^2/2}$. Coefficients were computed by 80-digit quadrature for even $k \le 60$; roots at 120 digits. Code: `xi_polyapprox.py`.

## 2 Coefficient decay: an inf-convolution

If the target were itself Gaussian-decaying (the setting of our Szegő-type paper), $\beta_k$ would decay like $3^{-k/2}$, the rate of the Hermite matching matrix. Measured local slopes of $\ln|\beta_k|$ per unit $k$ are

$$-0.24,\ -0.21,\ -0.22,\ -0.23,\ -0.25,\ -0.26 \qquad (k = 4, 14, 24, 34, 44, 54)\ :$$

far from $-\frac{1}{2}\ln 3 = -0.549$, and slowly steepening. The mechanism: inserting $F = \sum_m b_m \psi_m$ into (1) gives $\beta_k = \sum_m b_m M_{mk}$ with the Gaussian overlap matrix $M_{mk} = \langle \psi_m e^{-t^2/2}, \psi_k \rangle \asymp 3^{-(k-m)/2}$. On the log scale this is an inf-convolution:

$$-\ln|\beta_k| \ \approx\ \min_m \Big( -\ln|b_m| + \tfrac{\ln 3}{2}(k-m) \Big).$$

The geometric rate $\frac{1}{2}\ln 3$ can only dominate once the local slope of $-\ln|b_m|$ exceeds it, which the section data locates near $m \approx 42$; below that the arithmetic decay of $b_m$ sets the (slower) pace. The slopes should therefore drift to $-0.549$ for $k$ well beyond 60 — a prediction we have not yet tested.

## 3 Zeros: three zones, a quadratic balance law

The tail of the polynomial approximation is a combination of the modes $\tilde{h}_k = \psi_k e^{t^2/2}$, which *grow* like the inverse square root of the weight. Spurious zeros appear where the tail overtakes the target:

$$|\beta_N|\, e^{t^2/2} \ \gtrsim\ |\Xi(\lambda t)| \asymp e^{-\pi\lambda t/4} \quad \Longleftrightarrow \quad \tfrac{t^2}{2} + \tfrac{\pi\lambda}{4}\, t \ =\ -\ln|\beta_N|, \tag{2}$$

a quadratic balance law, in contrast to the sections' linear one $\frac{\pi}{4}T_c = -\ln|b_N|$ (up to the $T^{7/4}$ factor). The extra $t^2/2$ on the left is the price of the basis: it caps the capture height far below the sections'.

*Observation* 3.1. For $N = 20, 40, 60$, with $T_c$ from (2) using the measured $\beta_N$:

| | $N = 20$ | $N = 40$ | $N = 60$ |
|---|---|---|---|
| predicted $T_c$ | 6.8 | 12.0 | 17.5 |
| Riemann zeros captured | 0 | 1 | 2 |
| first detached real zero | 15.44 | 21.71 | 26.16 |
| sections (companion paper) | $\approx 1$ | $\approx 4$ | $\approx 7$ |

At $N = 60$ the approximant reproduces 14.1348 and 21.0293 (true: 14.1347, 21.0220); the third real zero sits at 26.16 against the true 25.01. Beyond the crossover the real zeros drift to the free spacing $\lambda\pi/\sqrt{2N} \approx 2.9$ of the top Hermite mode and terminate at 102.8, near the turning point $\lambda\sqrt{2N} \approx 109.5$. The only non-real zeros — $1, 2, 3$ conjugate pairs for $N = 20, 40, 60$ — again sit at the crossover, but now form short arcs climbing off the axis: at $N = 60$, $(24.5, 1.7)$, $(21.6, 8.1)$, $(17.2, 16.4)$ in $(T, \mathrm{Im})$ coordinates.

## 4 The updated taxonomy

| target / basis | decay | balance law | spurious zeros |
|---|---|---|---|
| $e^z$ / monomials (Szegő) | $1/k!$ | — | complex curve only |
| $\psi_m$ / Hermite polys | $3^{-k/2}$ | $t^2 = \frac{N}{2}\ln 3$ | real band $[\alpha, 1]$ + arcs |
| $\Xi$ / Hermite fns | superexp. in $\sqrt{k}$ | linear $(2)'$ | real past $T_c$, defects at $T_c$ |
| $\Xi$ / Hermite polys | inf-convolution | quadratic (2) | real past $T_c$, defect *arcs* |

The moral, stated multiplicatively: to see Riemann zeros through truncations on $\mathbb{R}_+$, the Gaussian factor $e^{-(\log x)^2/2}$ in the log-Hermite *functions* is not cosmetic. Dropping it — approximating by $x^{-1/2}$ times bare polynomials in $\log x$ — trades the linear balance law for a quadratic one and shrinks the verified height from $T \approx 46$ to $T \approx 21$ at degree 60. Both pictures obey the same principle: spurious zeros appear exactly where the truncation tail overtakes the target, and complex zeros mark the boundary. Only the exponents change.